CBSE Test Paper 05

Chapter 3 Motion in A Straight Line

1. Jules Verne in 1865 proposed sending people to the Moon by firing a space capsule from 220-m-long cannon with a final velocity of 10.97 km/s. What would have been the unrealistically large acceleration experienced by the space travelers during launch? **1**
    a. $2.74 \times 10^6\ m/s^2$
    b. $2.74 \times 10^5\ m/s^2$
    c. $3.74 \times 10^5\ m/s^2$
    d. $2.74 \times 10^3\ m/s^2$
2. Path length is a **1**
    a. tensor
    b. Derived unit
    c. scalar
    d. vector
3. Motion along a straight line is called __________. **1**
    a. parabolic motion
    b. circular motion
    c. oscillatory motion
    d. rectilinear motion
4. A truck accelerates at 1 m / sec^2 from rest. What is its velocity in m/s at a time of 2 sec? **1**
    a. 2
    b. 4
    c. 1
    d. 3
5. A stone thrown from the top of a building is given an initial velocity of 20.0 m/s

straight upward. Determine the time in seconds at which the stone reaches its maximum height. g =9.8 m / sec^2. **1**

   a. 2.8
   b. 2.04
   c. 1.67
   d. 2.7

6. For which condition, the magnitude of average velocity is equal to the average speed for a particular motion? **1**

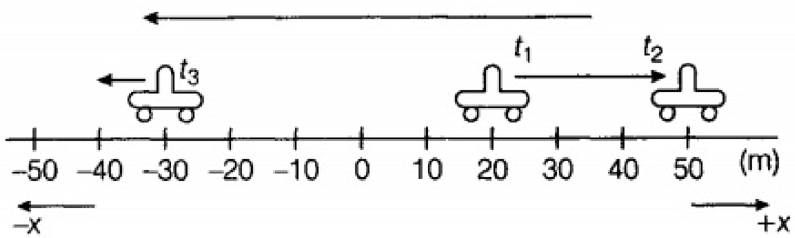
7. If the position of a particle at instant t is given by $x = 2t^3$, find the acceleration of the particle. **1**

8. The displacement - time graph of a particle is parallel to time - axis, what is the velocity of the particle? **1**

9. The displacement x of a particle moving in one dimension under the action of constant force is related to the time by the equation where x is in meters and t is in seconds. Find the velocity of the particle at **2**

   i. t = 3s
   ii. t = 6s.

10. For the motion shown in the figure, find the displacement of car between the time intervals t_1 and t_3. **2**



11. The displacement (in metre) of a particle moving along x-axis is given by $x = 18t + 15t^2$. Find the instantaneous velocity at t = 0 and t = 2 s. **2**

12. Two trains, each having a speed of 30 km/h, are headed towards each other on the same track. A bird that can fly 60 km/h flies off the front of one train when they were

60 km apart and heads directly to the other train. On reaching the train, the bird flies back to the first train.

What is the total distance the bird travels before the trains collide? **3**

13. A train takes 4 min to go between stations 2.25 km apart starting and finishing at rest. The acceleration is uniform for the first 40 s and the deceleration is uniform for the last 20 s.

    Assuming the velocity to be constant for the remaining time, calculate the maximum speed, acceleration, and retardation, use the only graphical method. **3**

14. A bullet bike moving on a straight road at a speed of 120 kmph is made to stop by a police officer within a 100m distance. Calculate the retardation of the bike (assumed uniform) and the time it takes for the bike to stop? **3**

15. Two towns A and B are connected by a regular bus service with a bus leaving in either direction every T minutes. A man cycling with a speed of 20 km h^{-1} in the direction A to B notices that a bus goes past him every 18 min in the direction of his motion, and every 6 min in the opposite direction. What is the period T of the bus service and with what speed (assumed constant) do the buses ply on the road? **5**